

6.2: Independence

Definition: We say two random variables X and Y are independent if

$$P(X \in A, Y \in B) = P(X \in A) \cdot P(Y \in B) \text{ for all } A, B \subseteq \mathbb{R}$$

In terms of joint distributions, this means $F(a, b) = F_X(a)F_Y(b)$ for all $a, b \in \mathbb{R}$

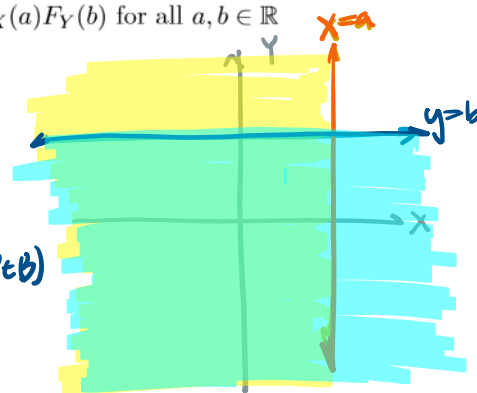
$$\text{LET } A = (-\infty, a] \rightarrow P(X \in A) = P(X \leq a) = F_X(a)$$

$$\text{LET } B = (-\infty, b] \rightarrow P(Y \in B) = P(Y \leq b) = F_Y(b)$$

$$P(X \in A, Y \in B) = P(X \leq a, Y \leq b) = F_{XY}(a, b)$$

$$\text{INDEPENDENT} \Leftrightarrow P(X \in A, Y \in B) = P(X \in A) \cdot P(Y \in B)$$

$$F_{XY} = F_X \cdot F_Y$$



If X and Y are both discrete, what does that look like for their point mass distributions?

$$\text{LET } A = \{a\} \rightarrow P(X \in A) = P(X = a) = p_X(a)$$

$$\text{LET } B = \{b\} \rightarrow P(Y \in B) = P(Y = b) = p_Y(b)$$

$$P(X \in A, Y \in B) = P(X = a, Y = b) = p_{XY}(a, b)$$

INDEPENDENT



$$p_{XY} = p_X \cdot p_Y$$

If X and Y are both continuous, what does that look like for their densities?

$$\text{LET } A = [a, a+\Delta x] \rightarrow P(X \in A) = \int_a^{a+\Delta x} \int_{-\infty}^{\infty} f_{XY}(x, y) dy dx = \int_a^{a+\Delta x} f_X(x) dx \xrightarrow{\Delta x \rightarrow 0} f_X(a) \cdot \Delta x$$

$f_X(x) = \text{marginal density of } x$

$$\text{LET } B = [b, b+\Delta y] \rightarrow P(Y \in B) = \int_b^{b+\Delta y} \int_{-\infty}^{\infty} f_{XY}(x, y) dx dy = \int_b^{b+\Delta y} f_Y(y) dy \xrightarrow{\Delta y \rightarrow 0} f_Y(b) \cdot \Delta y$$

$f_Y(y) = \text{marginal density of } y$

$$P(X \in A, Y \in B) = \int_a^{a+\Delta x} \int_b^{b+\Delta y} f_{XY}(x, y) dy dx$$

SS

$$\Delta x \rightarrow 0$$

$$\Delta y \rightarrow 0$$

$$f_{XY}(a, b) \cdot \Delta x \Delta y$$

$$P(X \in A, Y \in B) = P(X \in A) \cdot P(Y \in B)$$

$$f_{XY} \cdot \Delta x \Delta y = f_X \cdot \Delta x \cdot f_Y \cdot \Delta y$$

$$f_{XY} = f_X \cdot f_Y$$

Example: If $X = \text{Bin}(n, p)$ and $Y = \text{Bin}(n, q)$ are independent random variables,
 find $p_{XY}(i, j)$ for each $0 \leq i, j \leq n$ and $P(X + Y = n)$

$$X = \text{Bin}(n, p) \rightarrow p_X(i) = \binom{n}{i} p^i (1-p)^{n-i} \text{ for } i = 0, 1, \dots, n$$

$$Y = \text{Bin}(n, q) \rightarrow p_Y(j) = \binom{n}{j} q^j (1-q)^{n-j} \text{ for } j = 0, 1, \dots, n$$

$$p_{XY}(i, j) = p_X(i) \cdot p_Y(j) = \binom{n}{i} p^i (1-p)^{n-i} \binom{n}{j} q^j (1-q)^{n-j}$$

for $i = 0, 1, \dots, n$
 for $j = 0, 1, \dots, n$

$$P(X+Y=n) = P\left(\bigcup_{i=0}^n \{X=i, Y=n-i\}\right)$$

$$= \sum_{i=0}^n P(X=i, Y=n-i)$$

$$= \sum_{i=0}^n p_{XY}(i, n-i)$$

$$= \sum_{i=0}^n \binom{n}{i} p^i (1-p)^{n-i} \cdot \binom{n}{n-i} q^{n-i} (1-q)^{n-(n-i)}$$

$$= \sum_{i=0}^n \binom{n}{i} \binom{n}{n-i} [p(1-q)]^i [(1-p)q]^{n-i}$$

Example: Let $p_{XY}(1,0) = \frac{2}{9}, p_{XY}(1,1) = \frac{1}{3}$.

If X and Y are independent and $X, Y \in \{0, 1\}$, then what is $E[(X+Y)^2]$?

	$X=0$	$X=1$
$Y=0$	$\frac{4}{9} \cdot \frac{2}{5} = \frac{8}{45}$	$\frac{2}{9}$
$Y=1$	$\frac{4}{9} \cdot \frac{3}{5} = \frac{4}{15}$	$\frac{1}{3}$

$$p_{XY}(a,b) = p_X(a) \cdot p_Y(b)$$

$$p(Y=0) = \frac{2}{5}$$

$$p_{XY}(1,0) = p_X(1) \cdot p_Y(0)$$

$$\frac{2}{9} = \frac{5}{9} \cdot \frac{2}{5} \checkmark$$

$$p(X=1) = \frac{2}{9} + \frac{1}{3} = \frac{5}{9} = p_X(1)$$

$$p(X=0) = 1 - p(X=1) = 1 - \frac{5}{9} = \frac{4}{9}$$

$$p(Y=1) = 1 - p(Y=0) = 1 - \frac{2}{5} = \frac{3}{5}$$

$$\begin{aligned}
 E[(X+Y)^2] &= (0+0)^2 \cdot p_{XY}(0,0) + (1+0)^2 \cdot p_{XY}(1,0) + (0+1)^2 \cdot p_{XY}(0,1) + (1+1)^2 \cdot p_{XY}(1,1) \\
 &= 0 \cdot \frac{8}{45} + 1 \cdot \frac{2}{9} + 1 \cdot \frac{4}{15} + 4 \cdot \frac{1}{3} \\
 &= \frac{82}{45}
 \end{aligned}$$

Example: A man and woman match on tinder and agree to meet "after five" at a local bar. They both independently intend to show up sometime between 5 and 6 pm (uniformly distributed), and will leave after 15 minutes if they do not see the other person. What are the chances that they will meet?

X = AMOUNT OF MINUTES AFTER 5 PM MAN SHOWS UP.

$$= \text{UNIF}(0, 60]$$

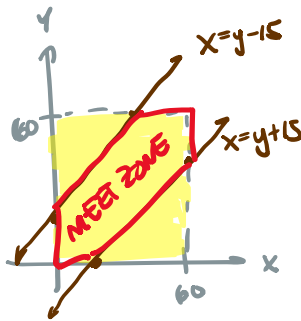
$$f_X(x) = \frac{1}{60-0} = \frac{1}{60} \quad 0 \leq x \leq 60$$

Y = AMOUNT OF MINUTES AFTER 5 PM WOMAN SHOWS UP.

$$= \text{UNIF}(0, 60]$$

$$f_Y(y) = \frac{1}{60-0} = \frac{1}{60} \quad 0 \leq y \leq 60$$

$$f_{XY}(x,y) = f_X(x) \cdot f_Y(y) = \frac{1}{60} \cdot \frac{1}{60} = \frac{1}{60^2} \quad \text{for } 0 \leq x \leq 60, 0 \leq y \leq 60$$



$$P(\text{MEET}) = P(|X-Y| \leq 15)$$

$$= P(-15 \leq X-Y \leq 15)$$

$$= P(Y-15 \leq X \leq Y+15)$$

$$= 1 - P(X \leq Y-15) - P(X \geq Y+15)$$

$$= 1 - \int_{15}^{60} \int_0^{y-15} \frac{1}{60^2} dx dy - \int_0^{45} \int_{y+15}^{60} \frac{1}{60^2} dx dy$$

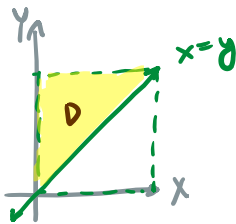
$$= 1 - \frac{1}{60^2} (\text{AREA OF TOP TRIANGLE}) - \frac{1}{60^2} (\text{AREA OF BOT TRIANGLE})$$

$$= 1 - \frac{1}{60^2} \cdot \frac{1}{2} \cdot 45 \cdot 45 - \frac{1}{60^2} \cdot \frac{1}{2} \cdot 45 \cdot 45$$

$$= 1 - \left(\frac{45}{60}\right)^2 = 1 - \left(\frac{3}{4}\right)^2 = \boxed{\frac{7}{16}}$$

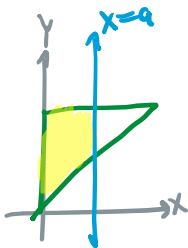
Example: Let $f_{XY}(s, t) = \begin{cases} kst^2 & 0 \leq s < t \leq 1 \\ 0 & \text{else} \end{cases} = \begin{cases} kxy^2 & 0 \leq x \leq y \leq 1 \\ 0 & \text{else} \end{cases}$

Are the random variables X and Y with this joint density independent?



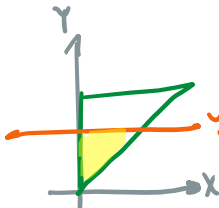
$$1 = \iint_D kxy^2 dA = k \int_0^1 \int_0^y xy^2 dx dy = k \int_0^1 \frac{1}{2} x^2 y^2 \Big|_0^y dy$$

$$= \frac{k}{2} \int_0^1 y^4 dy = \frac{k}{10} y^5 \Big|_0^1 = \frac{k}{10} \rightarrow \underline{k=10}$$



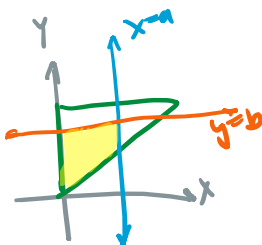
$$F_X(a) = \int_0^a \int_x^1 10xy^2 dy dx = \int_0^a \frac{10}{3} xy^3 \Big|_x^1 dx = \frac{10}{3} \int_0^a x - x^4 dx$$

$$= \frac{10}{3} \cdot \left(\frac{1}{2} x^2 - \frac{1}{5} x^5 \right) \Big|_0^a = \frac{10}{3} \cdot \left(\frac{a^2}{2} - \frac{a^5}{5} \right) = \underline{\underline{\frac{5a^2 - 2a^5}{3}}}$$



$$F_Y(b) = \int_0^b \int_0^y 10xy^2 dx dy = \int_0^b 5x^2 y^2 \Big|_0^y dy = \int_0^b 5y^4 dy$$

$$= y^5 \Big|_0^b = \underline{\underline{b^5}}$$



$$F_{XY}(a, b) = \int_0^a \int_x^b 10xy^2 dy dx = 10 \int_0^a \frac{1}{3} xy^3 \Big|_x^b dx$$

$$= \frac{10}{3} \int_0^a b^3 x - x^4 dx = \frac{10}{3} \left(\frac{1}{2} b^3 x^2 - \frac{1}{5} x^5 \right) \Big|_0^a$$

$$= \frac{10}{3} \cdot \left(\frac{b^3 a^2}{2} - \frac{a^5}{5} \right) = \underline{\underline{\frac{5b^3 a^2 - 2a^5}{3}}}$$

INDEPENDENT?

$$F_{XY} \stackrel{?}{=} F_X \cdot F_Y$$

$$\frac{5b^3 a^2 - 2a^5}{3} \neq \frac{5a^2 - 2a^5}{3} \cdot b^5$$

NOT INDEPENDENT